Replication of the social information transmission findings by Mesoudi, Whiten & Dunbar (2006, British Journal of Psychology)

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Introduction

Mesoudi, Whiten, and Dunbar (2006) asked whether some kinds of information survive transmission better than others. They used the transmission chain method, in which one participant's recall becomes the subsequent participant's stimulus, so that information fidelity and loss can be tracked as it travels. In Experiment 2, the first participant in a four-person chain read four short paragraphs, wrote down what they remembered, and then that written recall

was given to the next participant in the chain. The four paragraphs matched in structure and were close in length, each containing 14 propositions across 5 sentences and written as a chain of events linked causally and temporally so that no paragraph was more coherent than another. However, the content type of these paragraphs differed: gossip about third-party social relationships, ordinary non-gossip social interactions, individual behavior, and physical events. Every generation of the chain's recall was scored against the original paragraph, using Kintsch (1974) propositional analysis. The target finding for this replication is that social information survived transmission better than non-social information, which the original showed through the effect of content type on how many original propositions were recalled correctly, $F(3, 24) = 12.99$. Gossip and non-gossip social information survived transmission better than individual behavior or physical events, and the contrast between gossip and non-gossip social information was null at $F(1, 9) = 2.40$, which the authors interpreted as being due to the presence of people interacting with each other rather than the scandalous nature of the gossip content.

My current research asks how descriptions of people change as they pass between people, analyzing the compression and content convergence of social information across generations of transmission. I chose to replicate Mesoudi et al.'s experiment because it is a simpler version of the study I am about to run. However, while Mesoudi et al. are interested in how differing content in general survives transmission, I am interested in how differing *social* content (relationships, traits, actions, locations) about a single person survives transmission when presented all at once in one recall. Their four paragraphs were designed to vary widely from non-social to social in content, so this replication tested whether a multi-generational chain is a sensitive enough instrument to detect content effects at all. If it could not separate narratives that differ this greatly, it would likely also not separate smaller differences in social content such as relationships and traits. I replicated Experiment 2 online with UCSD undergraduates from SONA, through a web application I built for my own chain study that passed each recall to the next participant (see Procedure). The main differences from the original were running the study online, scoring recalls with LLMs, and setting the sample size by simulation. Because online participants may be less attentive or use outside help, the web application added a minimum reading time, attention checks, and screening for AI-assisted recalls. Scoring every recall by hand was not feasible at scale, so recalls were scored by LLMs, which Martinez (2024) found to reach an r of 0.86 to 0.94 against manual scoring of story recall, and validated against my hand coding of a subset. Finally, because the original reported no power analysis, the number of chains came from a simulation. Each is described in Methods.

Links

Repository: https://github.com/kyaneos/mesoudi2006
Original paper: 2006_mesoudi_social_information_bias.pdf (Retrieved on Oct 12, 2026 from doi.org/10.1348/000712605X85871)

Methods

Power Analysis

One row of data

Each row of my data is one participant's recall of one paragraph. It records the chain, the participant's generation, which material the paragraph was, the paragraph's position in the chain's original reading order (1 to 4), and the score. The score is the number of the paragraph's 14 original propositions the participant recalled correctly, scored against the original (F0) paragraph at every generation, as in Mesoudi et al. (2006). Each participant contributes four rows, so 12 chains of four generations give 192 rows.

chain	generation	material	position	score
3	2	social	3	9

How the simulated data is made

Because the chain is the unit of analysis, the simulation skips individual recalls and simulates the one number per chain that the planned test sees, the chain's social minus non-social difference (defined below). A simulated study draws one difference per chain from a normal distribution with the SD of Mesoudi's 10 real differences and a mean set to a share of his mean difference. Each of the four pairwise differences is simulated the same way.

The mean and SD come from the raw data for Experiment 2, which Mesoudi has since published (github.com/amesoudi/mesoudi_whiten_dunbar_2006). His `correct_recall` column is the same measure as my `score`, so I rename it.

All of the code in this section is in `new_code/power/power_sim.R` and runs live in this report.

```
library(tidyverse)
set.seed(240) # the same random numbers on every render

# SciMinds colors: a blue ramp, darker = bigger effect
sciminds_ramp <- c("0.5" = "#A6D0EA", "0.75" = "#58A6D6", "1" = "#2A7CBE")
# monospace figure text, and a lighter panel so the palest blue stays visible
theme_set(theme_gray(base_family = "mono") +
            theme(panel.background = element_rect(fill = "grey97")))

# Mesoudi's Experiment 2 data, pinned to one version so it never changes
mesoudi_url <- paste0(
  "https://raw.githubusercontent.com/",
  "amesoudi/mesoudi_whiten_dunbar_2006/",
```

```
  "a27a5c3024d55c47d17b30e6cdea7dd28c24e9fc/",
  "mesoudi2006_exp2_data.csv")
mesoudi <- read_csv(mesoudi_url, show_col_types = FALSE) |>
  rename(score = correct_recall) # propositions recalled correctly, out of 14
```

The simulation also adds noise for an automated scorer that agrees with human coding at r = 0.90, about what such a scorer would add to a chain's difference once its errors on the 16 paragraph scores are averaged. If human coders agree with each other at less than 0.90, a scorer that passes could be noisier, and the power reported here would be somewhat optimistic. A scorer biased toward one material is not modeled, because the validation checks for that separately (see Analysis Plan).

The planned test

Because the chain is the unit, the test sees one number per chain. This number is the chain's average score on the social materials (gossip and social) minus its average score on the non-social materials (individual and physical), across all four generations. Every score is already measured against the original paragraph, so a positive difference means more of the original social material survived the chain. This matches Mesoudi's material effect, which is also averaged over generations.

```
# each chain's social minus non-social difference, plus the four pairs
chain_difference <- function(data) {
  data |>
    group_by(chain, material) |>
    summarize(score = mean(score), .groups = "drop") |>
    pivot_wider(names_from = material, values_from = score) |>
    mutate(difference = (gossip + social) / 2 - (individual + physical) / 2,
           gossip_vs_individual = gossip - individual,
           gossip_vs_physical = gossip - physical,
           social_vs_individual = social - individual,
           social_vs_physical = social - physical)
}

real_differences <- chain_difference(mesoudi)$difference
```

The test is a sign-flip permutation test. If social and non-social information are transmitted equally well, each chain's difference is just noise, and it is as likely to be negative as positive. Flipping each chain's sign at random 5,000 times shows what the mean difference looks like when there is no effect. The p-value is the share of flipped means at least as far from zero as the real mean.

```r
sign_flip_p <- function(differences, n_flips = 5000) {
  n <- length(differences)
  # each row is one round of random signs, a + or - for every chain
  signs <- matrix(sample(c(-1, 1), n_flips * n, replace = TRUE),
                  nrow = n_flips)
  # multiplying gives every round's mean difference at once, which is much
  # faster than a loop that flips one round at a time
  flipped_means <- (signs %*% differences) / n
  # how often a flipped mean is as far from zero as the real one; the tiny
  # 1e-9 allowance keeps rounding error from missing exact ties
  mean(abs(flipped_means) >= abs(mean(differences)) - 1e-9)
}

real_p <- sign_flip_p(real_differences)
```

All 10 of Mesoudi's per-chain differences are positive, so the exact p is 2/1,024 ≈ 0.002, and the 5,000 random flips estimate it at 0.002.

Because Mesoudi's effect comes from only 10 chains, it is a single draw, and the true effect could be smaller. To see how much smaller is plausible, I bootstrapped his 10 per-chain differences, resampling them with replacement 10,000 times and taking the mean each time.

```r
boot_means <- map_dbl(1:10000,
                      \(i) mean(sample(real_differences, replace = TRUE)))

# the range of plausible effects
boot_ci <- quantile(boot_means, c(0.025, 0.975))
# the low end as a share of his effect
boot_low_share <- boot_ci[1] / mean(real_differences)
```

The 95% interval runs from 2.06 to 3.78 propositions, around his observed 2.85. Its low end is about 72% of his effect. This is why I plan around 75% of his effect, with 50% as a more pessimistic case. With only 10 chains, the interval itself is rough.

Power

The first table below gives Mesoudi's mean and SD for each test. For every combination of number of chains and true effect, I simulated 1,000 studies and ran the tests on each. The number of chains ranges from 4 to 48 in multiples of 4, so the reading orders balance, and the true effect is 100%, 75%, or 50% of Mesoudi's. Power is the share of studies that came out significant, at $p < 0.05$ for the planned test and at $p < 0.01$ for the four pairwise comparisons,

which is Mesoudi's Bonferroni-corrected level. The gossip versus social comparison is left out, because the original found no difference to power for.

```
# readable test names, in the order the tables and figure show them
test_labels <- c(difference = "social vs non-social",
                 gossip_vs_individual = "gossip vs individual",
                 gossip_vs_physical = "gossip vs physical",
                 social_vs_individual = "social vs individual",
                 social_vs_physical = "social vs physical")

# Mesoudi's effect for each test: the mean and SD of his 10 chain
# differences, and the cutoff it is tested at (0.01 for the pairs, his
# Bonferroni level)
mesoudi_effects <- chain_difference(mesoudi) |>
  pivot_longer(difference:social_vs_physical,
               names_to = "test", values_to = "chain_difference") |>
  mutate(test = factor(test, levels = names(test_labels),
                       labels = test_labels)) |>
  group_by(test) |>
  summarize(mean = mean(chain_difference), sd = sd(chain_difference)) |>
  mutate(alpha = if_else(test == "social vs non-social", 0.05, 0.01))

knitr::kable(mesoudi_effects, digits = 2,
             col.names = c("test", "mean difference", "SD", "cutoff"))
```

test	mean difference	SD	cutoff
social vs non-social	2.85	1.44	0.05
gossip vs individual	3.35	1.59	0.01
gossip vs physical	3.58	2.62	0.01
social vs individual	2.12	2.11	0.01
social vs physical	2.35	2.32	0.01

```
# scorer noise, as a share of the true SD, that makes measured and true
# differences correlate at r = 0.90
scorer_noise <- sqrt(1 / 0.90^2 - 1)

# one simulated study with n_chains chains: a p-value for each test. Each
# test gets its own simulated chains, which doesn't change any one test's
# power
one_experiment <- function(n_chains, effect) {
  map_dbl(1:nrow(mesoudi_effects), \(i) {
```

```
    this_mean <- effect * mesoudi_effects$mean[i]
    this_sd <- mesoudi_effects$sd[i]
    true <- rnorm(n_chains, mean = this_mean, sd = this_sd)
    measured <- true + rnorm(n_chains, mean = 0, sd = this_sd * scorer_noise)
    # 1,000 flips keeps the render to about a minute
    sign_flip_p(measured, n_flips = 1000)
  })
}

# power: the share of simulated studies that come out significant, for
# each test
estimate_power <- function(n_chains, effect, n_sims = 1000) {
  # one row per test, one column per simulated study, so each row is
  # compared with its own test's cutoff
  p_values <- replicate(n_sims, one_experiment(n_chains, effect))
  tibble(test = mesoudi_effects$test,
         power = rowMeans(p_values < mesoudi_effects$alpha))
}
```

```
# with no effect, each test should come out significant at about its cutoff
estimate_power(n_chains = 12, effect = 0, n_sims = 2000) |>
  mutate(cutoff = mesoudi_effects$alpha) |>
  knitr::kable(digits = 3,
               col.names = c("test", "share significant", "cutoff"))
```

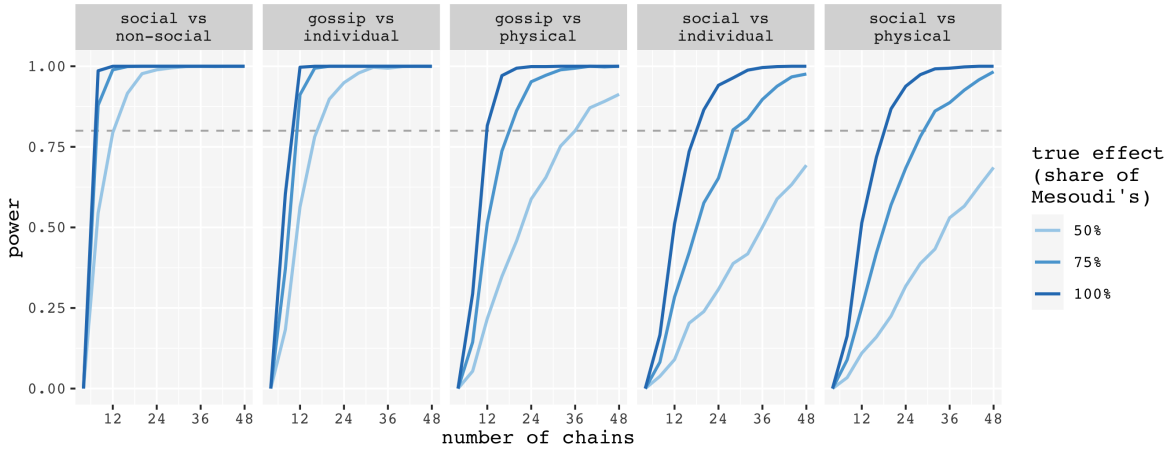
test	share significant	cutoff
social vs non-social	0.049	0.05
gossip vs individual	0.007	0.01
gossip vs physical	0.009	0.01
social vs individual	0.007	0.01
social vs physical	0.013	0.01

As a check, I ran the same simulation with no effect at all. Each test should then come out significant only as often as its cutoff allows, about 5% of the time for the planned test and 1% for the pairs, and it does.

```
# 4 to 48 chains in steps of 4, so the reading orders balance
power <- expand_grid(n_chains = seq(4, 48, by = 4),
                     effect = c(1, 0.75, 0.5)) |>
```

```
  mutate(result = map2(n_chains, effect, \(n, e) estimate_power(n, e))) |>
  unnest(result)
```

```
ggplot(power, aes(x = n_chains, y = power, color = factor(effect))) +
  geom_hline(yintercept = 0.8, linetype = "dashed", color = "grey70") +
  geom_line(linewidth = 0.8) +
  facet_wrap(~test, nrow = 1, labeller = label_wrap_gen(width = 10)) +
  scale_x_continuous(breaks = c(12, 24, 36, 48)) +
  scale_color_manual(values = sciminds_ramp,
                     labels = c("50%", "75%", "100%")) +
  labs(x = "number of chains", y = "power",
       color = "true effect\n(share of\nMesoudi's)")
```



Each panel is one test, and the first panel is the planned test. Each line is one true effect size, with darker lines for bigger effects, and the dashed line marks 80% power.

The table below gives the smallest number of chains that reaches 80%, 90%, and 95% power, and the power at 12 chains.

```
# the fewest chains that reach a power target, or "> 48" if none do
chains_for <- function(n_chains, power, target) {
  if (!any(power >= target)) return("> 48")
  as.character(min(n_chains[power >= target]))
}

# the fewest chains for 80%, 90% and 95% power, and the power at 12 chains
chains_table <- power |>
  group_by(test, effect) |>
```

```
  summarize(chains_for_80 = chains_for(n_chains, power, 0.80),
            chains_for_90 = chains_for(n_chains, power, 0.90),
            chains_for_95 = chains_for(n_chains, power, 0.95),
            power_at_12 = power[n_chains == 12],
            .groups = "drop") |>
  arrange(test, desc(effect))

knitr::kable(chains_table, digits = 2,
             col.names = c("test", "share of Mesoudi's effect",
                           "chains for 80% power", "chains for 90% power",
                           "chains for 95% power", "power at 12 chains"))
```

test	share of Mesoudi's effect	chains for 80% power	chains for 90% power	chains for 95% power	power at 12 chains
social vs non-social	1.00	8	8	8	1.00
social vs non-social	0.75	8	12	12	0.99
social vs non-social	0.50	16	16	20	0.79
gossip vs individual	1.00	12	12	12	1.00
gossip vs individual	0.75	12	12	16	0.91
gossip vs individual	0.50	20	24	28	0.56
gossip vs physical	1.00	12	16	16	0.82
gossip vs physical	0.75	20	24	24	0.51
gossip vs physical	0.50	36	48	> 48	0.22
social vs individual	1.00	20	24	28	0.51
social vs individual	0.75	28	40	44	0.28
social vs individual	0.50	> 48	> 48	> 48	0.09
social vs physical	1.00	20	24	28	0.51

test	share of Mesoudi's effect	chains for 80% power	chains for 90% power	chains for 95% power	power at 12 chains
social vs physical	0.75	32	40	44	0.25
social vs physical	0.50	> 48	> 48	> 48	0.11

```
# planned test at 12 chains and 75% of the effect
power_at_12 <- power |>
  filter(test == "social vs non-social", effect == 0.75, n_chains == 12) |>
  pull(power)

# and at 50% of the effect
power_at_12_half <- power |>
  filter(test == "social vs non-social", effect == 0.5, n_chains == 12) |>
  pull(power)
```

The original study ran 10 chains and found that social material beat non-social material by 2.85 propositions per chain. Because an effect from 10 chains is likely inflated, and the low end of its bootstrapped interval is about 72% of the observed effect, I planned for 75% of it, which 8 chains detect with 80% power. Because the paragraph orders only balance at multiples of four, and I did not want to run fewer chains than the original, I took the first multiple of four above its 10 chains. I preregistered 12 chains, which gives 0.99 power for the planned test at 75% of the effect and 0.79 at half of it, with a scorer that agrees with human coding at $r = 0.90$. Chains kept opening up to a cap of 32, the point at which all four pairwise comparisons reach 80% power at 75% of the effect. Twelve chains was also feasible to fill on SONA within the collection window, and the cap of 32 kept a larger sample possible if sign-ups allowed. A significant result at 12 chains would replicate the social bias. A non-significant one would argue against an effect three quarters the size of Mesoudi's, though it could not rule out an effect much below half his size. Any pair that was not significant at the final sample would be reported as underpowered.

Planned Sample

I recruited UCSD undergraduates through the SONA participant pool for course credit. The only screening criterion was native or fluent English, and the web application ran only on a laptop or desktop computer. Based on the power analysis, I preregistered 12 chains (48 participants), with a cap of 32 chains (128 participants). Chains continued to open as participants signed up for one week after the study opened on SONA, or until November 25 if that came first, and collection ended four days after new chains stopped opening. Because I did not look at the

results during collection, these dates alone determined the final sample. If fewer than 12 chains were complete when collection ended, I would analyze the chains that were complete and use the same simulation to report the power that number gives.

Materials

The materials are the four paragraphs reproduced in Appendix B of Mesoudi et al. (2006), which I used verbatim. Each paragraph was "designed to constitute a comparable linear chain of events linked causally and temporally." This was meant to keep any one paragraph from being easier to remember simply because it told a more coherent story. The gossip paragraph, for example, reads: "Nancy is having an affair with her married college professor. Nancy recently became pregnant with the professor's child. The professor promised Nancy that he would leave his wife, but since Nancy told him she was pregnant, the professor refused to see her. So Nancy told the professor's wife about the affair. The professor's wife was so upset that she left the professor." As in the original, "the bracketed labels, indicating the type of material and the numbers of words, sentences and propositions, were not presented to the participants." The original states that order "was now counterbalanced," but does not say how. I therefore rotated chains through four orders from a balanced Latin square, in which each paragraph appears once in each position and follows every other paragraph once. Mesoudi also published a propositional breakdown of the four paragraphs alongside his data, which divides each paragraph into its 14 propositions. This breakdown, with each proposition restated in plain words for coders, served as the coding sheet for scoring.

Procedure

Because the design of Experiment 2 was "largely identical to that of Experiment 1," I followed the procedure described for Experiment 1: "A booklet was produced that contained on the first page the instruction, 'please read the following text through once. When you have finished turn the page', followed by the material. The second page contained the instruction, 'now, without turning back, please write out as best you can the text you just read. Be as accurate as possible, but don't worry if you can't remember it all. When you have finished turn the page', followed by a blank space for recall. A final sheet solicited the age and gender of the participant, and thanked them for participating, whereupon the experimenter debriefed them as to the nature of the study." As in the original, "no distractor task was performed, and no time limit was set. At no point in the procedure did either the printed instructions or the experimenter state that the experiment was a memory test, that the material had come from another participant, or that their recall would be passed on to another participant."

In my version, the booklet was a web application I built, which showed each page as its own screen, kept the original instructions word for word, collected each recall, and passed it to the next participant in the chain. Before the task, participants answered a screening question on English fluency and an instructions check that repeated until they passed it. The text screen

then held them for a minimum reading time, set at 400 words per minute for the length of their text. This is faster than a typical reading pace, so the minimum stopped skimming without holding up a normal reader. The text could not be selected or copied. Because participants could not physically turn the page back, the text disappeared before the recall screen appeared. On the recall screen, pasting was blocked, and the application logged how each recall was typed and whether the participant left the page. Both screens also carried instructions that participants could not see, which directed an AI to use an uncommon word in its response, so that AI-assisted recalls could be flagged. After recalling, participants answered an honesty item about outside help and an instructed-response item alongside the age and gender questions. They then read the debrief and returned to SONA for credit. In the original, the experimenter typed each recall into the next participant's booklet. My application passed each recall on only once it cleared the exclusion checks in the Analysis Plan, automatically correcting its spacing and nothing else. Participants received credit whether or not their recall was excluded.

Analysis Plan

I kept the original's two analysis rules. First, "each chain is treated as an independent unit of analysis, rather than each participant," because "the recalls of each participant within a single chain are not statistically independent of one another." Second, "recall in each generation was always compared with the original (F0) material, not the previous generation's recall." This means every recall was scored as the number of a paragraph's 14 original propositions it contains, giving four scores per participant and sixteen per chain.

Key analysis. The confirmatory test is the social versus non-social contrast. For each chain, I averaged its scores on the gossip and social paragraphs across all four generations, did the same for the individual and physical paragraphs, and took the difference. I then ran a two-sided sign-flip permutation test (described in the Power Analysis) on these per-chain differences, with 5,000 flips at $\alpha = 0.05$. The effect would count as replicated if the mean difference was positive and $p < 0.05$.

Additionally, I report three secondary analyses: 1) the original 4 (material) x 4 (generation) repeated-measures ANOVA, with chains as the units, without gender, and with Greenhouse-Geisser correction, 2) Mesoudi's own multilevel reanalysis, published with his data, and 3) the original's four social versus non-social pairwise comparisons, each run as the same sign-flip test on that pair's per-chain differences, at the original's Bonferroni-corrected $\alpha = 0.01$. Without gender, the ANOVA's material effect becomes $F(3, 27)$, and on Mesoudi's data it gives 11.81 against the published $F(3, 24) = 12.99$, which included gender. The gossip versus social comparison is reported descriptively, because a non-significant difference in this sample would not by itself replicate the original null.

Recalls were scored by two LLMs at temperature 0, Claude Opus 4.6 and a local Gemma 4 model, with the exact model versions recorded in the frozen scoring code. Each checked whether each of a paragraph's 14 propositions was present. Because Claude's output can vary

between runs even at temperature 0, it scored each recall three times, with the propositions in a different random order each time, and its score for each proposition was the majority of the three. Gemma gives the same output every time at temperature 0, so it scored each recall once. Because a scoring prompt can be adjusted until it gives a desired answer, the prompt was developed on pilot chains only, then frozen along with both model versions before any main data were scored. Whichever model agreed better with my hand coding on the pilot chains was the primary scorer, and the other served as a sensitivity check.

I hand coded about a third of the chains using Mesoudi's propositional breakdown. Human coders made the same judgment as the models, marking each of the 56 propositions present or absent in any wording. The original first divided each recall into its own propositions, which gave a measure of recall quantity. Because my confirmatory test only uses recall accuracy, I skipped this step. Because I knew the hypothesis, I coded these recalls in shuffled order with the chain and generation hidden. The material could not be hidden, since the text itself gives it away. Additionally, if a second coder was available, they would code a subset of the same chains blind to the hypothesis, and I would report our agreement alongside the original's.

Human coding is the standard, so automated scoring passed only if it agreed with my hand coding at least as well as two human coders agree with each other:

- Each model's paragraph totals had to correlate with mine at least as highly as the two human coders' totals correlated, or at a Pearson r of at least 0.90 if no second coder was available, with an ICC lower 95% bound of at least 0.75.
- Because a scorer that credits one kind of material more readily than another would bias the confirmatory test directly, I computed each hand-coded chain's difference from the model's scores and from mine. The mean absolute gap between them had to be no larger than the gap between the two human coders, or 0.5 propositions if no second coder was available.

If either condition failed, or if the two models disagreed on the confirmatory result, meaning their mean differences had opposite signs or only one was significant, I would hand code every recall. I also report Claude's agreement with itself across its three runs, the agreement between the two models, and Cohen's kappa for single propositions. All human and model scores are released with the analysis code.

Exclusions targeted only recalls that did not reflect the task. A poor recall is still a result, and excluding it would inflate how much appears to survive. Clear failures were excluded, and that generation of the chain was rerun with a new participant. These were recalls that contained the word requested by the hidden instructions, came from a participant who reported outside help on the honesty item, or were inserted rather than typed according to the typing log. The rules that needed a cutoff had fixed thresholds, set from the pilot data before collection began. A recall typed with fewer keystrokes than half its characters, or with any single jump of more than 20 characters, was inserted rather than typed. Borderline cases, in which the recall was under 10 words and under a quarter of the words in the text it was given, shared less than 15% of its four-letter-or-longer words with that text, or came from a participant who failed the

instructed-response item, held the chain until I reviewed them. For each, I only judged whether the recall was an attempt to recall the text. Leaving the page was logged, and a participant who was away for more than 10 minutes was timed out, and that generation was rerun.

Differences from Original Study

The original sample was 40 students who "participated voluntarily, were unpaid and had normal reading and writing ability," run in person "in groups of between one and five." My sample was UCSD undergraduates from SONA, participating alone and online for course credit. The number of chains was set by the power simulation, and the four counterbalanced orders balance when the number of complete chains is a multiple of four. Because collection ended on a date, some orders may have ended up with one more chain than others, and I report the number of chains per order. The original also balanced chains by gender, with five female and five male chains. I collected gender but did not balance or model it. Neither experiment found a gender effect (Exp 2: $F(1, 8) = 0.08$), and single-gender chains would be slow to fill on SONA, so this difference should not affect the result.

The task itself moved from a paper booklet to a web application. In the booklet, rereading, copying, and outside help were ruled out physically. The added measures, including an instructions screen with a check that participants understood to read the text once, the minimum reading time, non-selectable text, paste blocking, typing logs, hidden AI instructions, and the honesty and instructed-response items, exist to rule out the same things online. Excluded generations were rerun, which the original never needed. Recalls were also passed on with spacing corrected automatically, which is the closest online equivalent to the original experimenter typing them up. The original does not say how order was counterbalanced, so the balanced Latin square is my own choice.

The largest analytic difference is scoring. The original hand coded every recall using Kintsch (1974) propositional analysis, and I scored with LLMs validated against my hand coding of a subset, in which coders marked each proposition present or absent directly instead of first dividing the recall into propositions. My confirmatory test also differs. I used a permutation test on the social versus non-social contrast, and the original used a mixed ANOVA with planned comparisons. However, I report the original analysis alongside it.

Two of these differences could matter. Online participants may be less attentive, which would lower recall overall. Because every chain contributes both social and non-social scores, lower overall recall should lower both scores together, though it could also shrink the difference between them in absolute terms, or affect some materials more than others. The exploratory comparison of first-generation recall with the original's checks for this. Automated scoring also adds error, which would weaken the effect. This is why the power simulation includes the scoring error of a scorer that agrees with human coding at $r = 0.90$.

Methods Addendum (Post Data Collection)

You can comment this section out prior to final report with data collection.

Actual Sample

Sample size, demographics, data exclusions based on rules spelled out in analysis plan

Differences from pre-data collection methods plan

Any differences from what was described as the original plan, or "none".

Results

Data preparation

Each recall was stored exactly as typed, and the version passed to the next participant, with only spacing corrected, was stored separately. All scoring used the original. Because recalls that failed an exclusion check were removed and their generation rerun during collection, the data contain only accepted recalls, including borderline recalls that I reviewed and accepted before they were passed on. Chains that had not completed all four generations by the end of collection were dropped. Each remaining recall was scored against the 56 propositions of the four original paragraphs, giving four scores per participant. About a third of the chains, chosen at random before scoring, were also hand coded for the validation checks in the Analysis Plan. The final data have one row per participant per paragraph, which records the chain, the generation, the material, the paragraph's position in the chain's original reading order, and the number of its propositions recalled correctly.

Responses are loaded from the study app's export, and because the pilot ran on the test link, only test rows are kept for the pilot and only real rows for the study.

```
# every row from the study app, saved by `npm run export` in whispers
export_path <- "../new_code/whispers/exports/whispers_responses.json"
# exports/ is gitignored, so a fresh clone has no export file at all
responses <- if (file.exists(export_path)) {
  jsonlite::read_json(export_path, simplifyVector = TRUE)
} else {
  list()
}
# the export is missing or empty before the pilot, which skips the data
# chunks below
has_data <- length(responses) > 0
```

```
# TRUE: the pilot (classmates on the test link). FALSE: the real study
pilot <- TRUE
if (has_data) responses <- filter(responses, test == pilot)
knitr::kable(tibble(rows = if (has_data) nrow(responses) else 0))
```

rows
8

```
# the hand codes: the coding Google Sheet downloaded as an .xlsx
# (File > Download > Microsoft Excel) into the same gitignored folder, and
# the key from coding_sheet.R that links each recall code to its chain.
# Until the coded sheet is there, the hand-codes chunk below is skipped
exports <- "../new_code/whispers/exports/"
which_data <- if (pilot) "pilot" else "study"
coded_path <- paste0(exports, "coded_", which_data, ".xlsx")
key_path <- paste0(exports, "coding_key_", which_data, ".csv")
has_codes <- file.exists(coded_path)
```

```
# how many rows ended in each status
knitr::kable(count(responses, status))
```

status	n
passed	8

```
# every exclusion was decided before the next generation read the recall:
# canary and not_typed at submit, outside_help after the final questions,
# and review for a held recall (near_empty, off_topic, instructed_response)
# I excluded after reading it
excluded <- count(filter(responses, status == "excluded"), exclude_reason)
if (nrow(excluded) == 0) excluded <- tibble(exclude_reason = "none", n = 0)
knitr::kable(excluded)
```

exclude_reason	n
none	0

```r
# only accepted recalls go on to scoring
recalls <- filter(responses, status == "passed")

# held recalls I passed after reading them, by why they were held
held <- count(tibble(flag = as.character(unlist(recalls$flags))), flag)
if (nrow(held) == 0) held <- tibble(flag = "none", n = 0)
knitr::kable(held)
```

flag	n
none	0

```r
# seconds each accepted participant spent reading the text and writing
# their recall: the median, the fastest and the slowest. Nobody is
# trimmed, since the read screen already sets a minimum reading time
recalls |>
  select(reading = read_ms, writing = recall_ms) |>
  pivot_longer(everything(), names_to = "screen", values_to = "ms") |>
  group_by(screen) |>
  summarize(median = median(ms) / 1000,
            min = min(ms) / 1000,
            max = max(ms) / 1000) |>
  knitr::kable(digits = 1)
```

screen	median	min	max
reading	17.5	7.4	88.4
writing	32.4	17.6	140.1

```r
# only chains with all four generations accepted are analyzed. The app
# allows one accepted recall per chain and generation, so four rows
# means all four
complete <- count(recalls, chain) |> filter(n == 4)
recalls <- filter(recalls, chain %in% complete$chain)
knitr::kable(tibble(complete_chains = nrow(complete)))
```

complete_chains
1

```
# the rubric tab turns each column header back into its material. The
# material is only written on each block's first row, so it's filled down
rubric <- readxl::read_excel(coded_path, "rubric") |> fill(material)
key <- read_csv(key_path, show_col_types = FALSE)

# every tab except the rubric is one coder: one row per recall, one column
# per proposition, 1 if present and blank if not
coders <- setdiff(readxl::excel_sheets(coded_path), "rubric")
hand_codes <- map(coders, \(coder) {
  mutate(readxl::read_excel(coded_path, coder), coder = coder)
}) |>
  list_rbind()

# each coder's score per recall and paragraph: how many of its 14
# propositions were marked present, with the recall's chain and generation
hand_scores <- hand_codes |>
  pivot_longer(any_of(rubric$point),
               names_to = "point", values_to = "present") |>
  left_join(select(rubric, point, material), by = "point") |>
  summarize(score = sum(present == 1, na.rm = TRUE),
            .by = c(coder, recall, material)) |>
  left_join(select(key, recall, chain, generation), by = "recall")

hand_scores |>
  pivot_wider(names_from = material, values_from = score) |>
  arrange(coder, recall) |>
  knitr::kable()
```

Confirmatory analysis

Mesoudi et al. (2006) conclude that "human cultural transmission is biased toward information concerning social interactions over equivalent non-social information." This is the main inference of the paper. The original tested it in two steps. First, an overall ANOVA asked whether the four materials differed at all in how much of them survived, $F(3, 24) = 12.99$. This test does not say which materials did better, so it would have been just as significant if the physical paragraph had survived best. Second, four pairwise comparisons, each pairing a social paragraph with a non-social one, showed the direction of the difference. Neither step alone tests the claim.

I tested the claim in one step, using each chain's social minus non-social difference and the two-sided sign-flip permutation test described in the Power Analysis, with 5,000 flips at $\alpha = 0.05$. The test assumes only that, with no effect, each chain's difference is symmetric around zero, so it does not depend on normality, which 12 chains are too few to check. Because it is a

single comparison of two things, social and non-social, it needs no sphericity correction and no correction for multiple comparisons. The original's α of 0.01 was the 0.05 split across five comparisons.

If the average difference is positive and the test is significant, the effect replicates. Additionally, I report the original ANOVA and pairwise comparisons alongside it, so my results can be compared directly with the original's, and I plot correct recall by generation and material next to the original's Figure 4.

Confirmatory test on simulated data

Until the data are collected, this section runs the confirmatory test on one simulated study. The study uses the planned condition from the power analysis, which is 12 chains at 75% of Mesoudi's effect, scored by a scorer that agrees with true scores at only r = 0.90. Because the effect is built into the simulation, the estimate should land near the built-in difference, which is 75% of Mesoudi's mean difference. The seed is fixed, and the draw is reported as is. Once the real data are in, `chain_difference()` will turn them into `study_differences`, and the rest of the code below will run unchanged.

```
# the built-in difference: 75% of Mesoudi's
built_in <- 0.75 * mean(real_differences)

# one simulated study at the planned condition, measured by the r = 0.90
# scorer
set.seed(2006)
true_differences <- rnorm(12, mean = built_in, sd = sd(real_differences))
study_differences <- true_differences +
  rnorm(12, mean = 0, sd = sd(real_differences) * scorer_noise)
study_p <- sign_flip_p(study_differences)

tibble(chains = 12,
       estimate = mean(study_differences),
       built_in = built_in,
       chains_positive = sum(study_differences > 0)) |>
  knitr::kable(digits = 2)
```

chains	estimate	built_in	chains_positive
12	2.08	2.14	11

The estimate is 2.08 propositions against a built-in 2.14, and the test gives p = 0.003. This is one study. The power curve in the Power Analysis shows how often a study like this one succeeds as the number of chains and the size of the effect change.

Exploratory analyses

I will run two exploratory analyses. First, I will test whether a paragraph's position in the original reading order affects how much of it survives, which the balanced Latin square separates from material. Second, I will compare first-generation recall in each material with the original's, to see how recall in an online sample compares with the in-person original.

Discussion

Summary of Replication Attempt

Open the discussion section with a paragraph summarizing the primary result from the confirmatory analysis and the assessment of whether it replicated, partially replicated, or failed to replicate the original result.

Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis, (b) assessment of the meaning of the replication (or not) - e.g., for a failure to replicate, are the differences between original and present study ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the replication attempt. None of these need to be long.